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Oefeningenreeks 1C nr. 26.4

Opgave: Bereken in $\mathbb{C}$:

$$\begin{aligned}\frac{6 \operatorname{cis} 80^\circ}{2 \operatorname{cis} 20^\circ} &= \frac{6}{2} \cdot \operatorname{cis}(80^\circ - 20^\circ) = 3 \cdot \operatorname{cis}(60^\circ) \\ &= 3 (\cos 60^\circ + i \sin 60^\circ) = 3 \left(\frac{1}{2} + i \cdot \frac{\sqrt{3}}{2} \right) = \frac{3}{2} + \frac{3\sqrt{3}}{2}i\end{aligned}$$

Antwoord:

$$\boxed{\frac{3}{2} + \frac{3\sqrt{3}}{2}i}$$

References